Moving a power of 10 up or down: the sign of the exponent changes

From the bottom to the top

$$\frac{6}{10^{3}} \rightarrow 6 \cdot 10^{-3}$$

10^{3} goes **up**: its exponent changes from **+3** to **−3**.

From the top to the bottom

$$4 \cdot 10^{-2} \rightarrow \frac{4}{10^{2}}$$

10^{-2} goes **down**: its exponent changes from **−2** to **+2**.

Useful to divide

$$\frac{10^{5}}{10^{-2}} \rightarrow 10^{5} \cdot 10^{2} = 10^{7}$$

Move it up, then **add** the exponents.

Only the **power of 10** moves and changes the sign of its exponent. The **number** in front stays where it is.